Earth's energy budget from space: a guide to the CERES explorer

Ocean module · app: joymonteiro.github.io/intro_climate_science/tutorials/ceres-energy-budget

This handout goes with the app "CERES energy budget explorer". Open the app next to it. It runs in any browser, including on a phone, and needs no installation. You need only what lecture O1 covered: the Earth gains energy as sunlight and loses it as infrared radiation, and the difference is what heats or cools the planet.

- **Sections 1–3** explain where the data come from, what they can and can't tell you, and why scientists use them.
- **Section 4** is a tour of the app's five views.
- **Section 5** has eight exercises. Each one asks you to predict something **before you look**, tells you exactly what to set **in the app**, lists **what to look for**, and then explains **why**. Write your prediction down before you read on. The exercises follow the app's views, from the top of the atmosphere down to the surface and then to El Niño.
- **Section 6** gives the data sources and further reading.

> **The point in two sentences.** The Earth takes in more sunlight than it gives back as infrared radiation in the tropics, and gives back more than it takes in near the poles, so the atmosphere and ocean must carry energy poleward. Overall the planet is gaining a little energy every year, and most of that ends up in the ocean.

1. What CERES is

CERES stands for **Clouds and the Earth's Radiant Energy System**. It is a set of identical instruments that NASA has flown on several satellites: Terra, Aqua and, more recently, NOAA-20. Each instrument is a **broadband scanning radiometer**. It sweeps across the Earth below and measures how much radiation comes up from each spot. On Terra and Aqua it has three channels: a **shortwave** channel for reflected sunlight, a **total** channel for all wavelengths, and a narrow infrared "window" channel. The **longwave**, the infrared emitted by the Earth, is the total minus the shortwave. The newer instrument on NOAA-20 measures the longwave in a channel of its own.

"Broadband" is the important word. Most satellite instruments measure a few narrow colours, which is good for spotting clouds or measuring temperature. CERES instead catches all the energy, at every wavelength, because energy is what the budget is about.

Why do we need satellites at all? The top of the atmosphere is the only place where the Earth's energy budget is simple: energy can cross it only as radiation. Sunlight comes in, and reflected sunlight and infrared go out. No ground station can see that boundary. A satellite looks down on it from above, and a few satellites together see every part of the globe at many times of day.

The app uses **monthly mean maps**. Each map gives, for every box of the latitude–longitude grid, the average flux over one calendar month, counting day and night. The source data are on a 1° grid. The app averages them onto a 2° grid (a grid of 90 × 180 cells), weighting each 1° box by its area, to keep the download small. The record runs from January 2010 to January 2025, which is 181 months.

2. What EBAF is, and its honest caveats

The product in the app is **CERES EBAF**, which stands for **Energy Balanced and Filled**.

"Filled" refers to the clear-sky maps. To know what clouds do, we need the fluxes for clear skies (see below), but in many places and months CERES never sees a cloud-free scene, and a clear-sky map built from CERES alone has holes. EBAF fills them, using the clear parts of scenes seen by a finer-resolution imager (MODIS) and

radiative-transfer calculations. Separately, the CERES satellites pass over each place only a few times a day, and the hours in between are filled in using geostationary weather satellites.

"Energy Balanced" needs more care. Each CERES instrument is very stable, so it detects *changes* very well, but its absolute calibration is uncertain by a few W m^{-2}. That is more than the planet's whole imbalance. So the EBAF team adjusts the shortwave and the longwave fluxes, each within its own uncertainty, so that the long-term global mean of the net flux matches the rate at which the ocean is measured to be gaining heat. In practice this means:

- the **size** of the global-mean imbalance is anchored to ocean measurements, not measured by CERES alone;
- the **patterns** (where the Earth gains and loses), and the **changes** from month to month and year to year, come from CERES itself.

Surface fluxes are computed, not measured. No satellite can measure the radiation at the ground directly, because the atmosphere is in the way. The surface fields in EBAF come from a radiative-transfer model: a program that calculates how radiation passes through the atmosphere. The model is fed clouds and aerosols seen by satellite instruments, and air temperature and humidity from a weather-forecast reanalysis. Its results are then nudged so that, at the top of the atmosphere, they agree with what CERES measured. The surface numbers are therefore less certain than the top-of-atmosphere numbers, and anything computed from them (the "column" in view 3) inherits that uncertainty.

Clear-sky fluxes and the cloud radiative effect. To ask what clouds do, we need to know what the fluxes would be *without* the clouds. EBAF provides **clear-sky** fluxes. The app uses EBAF's **clear-sky for the total region** (the variables called `clr_t`). For each grid box and month, this is the flux that the *whole* box would have if its clouds were removed. Over the clear parts of a box it comes from measurements. Over the cloudy parts it comes from the radiative-transfer model with the clouds taken out. This is the same definition that climate models use, so the two can be compared.

The **cloud radiative effect** (CRE), called "Cloud effect" in the app, is then

```
cloud effect = (net flux, all skies) − (net flux, clear skies)
```

computed separately for the shortwave, the longwave and their sum. With the app's sign convention, a **negative** cloud effect means the clouds make the Earth (or the surface) **lose** energy compared with clear skies; a **positive** one means they make it **gain** energy.

3. Why it's used

- **It is the benchmark for climate models.** A climate model has to get the top-of-atmosphere budget right, both the global numbers and the maps. CERES is what modellers compare against.
- **Clouds.** How clouds will change as the planet warms is the largest source of uncertainty in how much warming a given amount of CO_2 will cause. Clouds both reflect sunlight and trap infrared. CERES measures both effects, everywhere, every month.
- **Monitoring Earth's energy imbalance.** CERES shows how the imbalance changes from year to year, and whether it is growing.
- **The link to the ocean.** Most of the energy the Earth gains ends up in the ocean, because water stores far more heat than air or land. The ocean is where the imbalance shows up as warming. EBAF's anchoring to ocean heat measurements is this same link, run in the other direction.

4. Reading the app

The app has five views, chosen with the numbered tabs at the top. Each view opens with a question in bold at the top. The question is there for you to think about; the app never answers it. The settings are your own: change any of them and look again.

Sign convention. Everywhere in the app, **positive means energy going into the thing named**: into the Earth at the top of the atmosphere (TOA), into the surface at the surface, and into the atmosphere for the column. Outgoing longwave (OLR) and incoming or absorbed sunlight are plain amounts, always positive; the field's note under the field menu says which is a gain and which a loss.

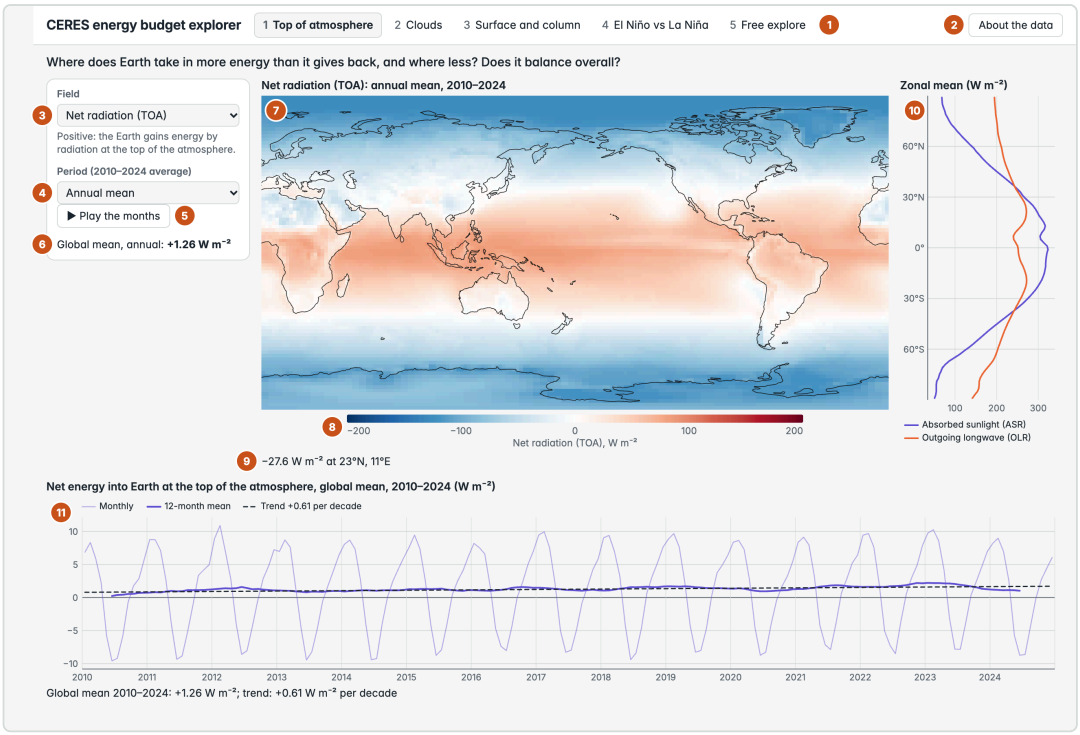
Mean and anomaly. A **mean** is the average of a field over the months you choose. An **anomaly** is the difference from the 2010–2024 average **of the same calendar months**. A January anomaly is measured against the average January, so the seasonal cycle drops out and what is left is how unusual that month was.

Seasons. DJF is December–January–February, MAM is March–May, JJA is June–August, SON is September–November. "DJF 2015–16" means December 2015 plus January and February 2016. In view 4, the **Jul–Jun year** runs from July of one year to June of the next, which matches the way El Niño events grow and decay. In views 1–3 the period is always the 2010–2024 average: either the **Annual mean** or one **average month** (for example, the average of all fifteen Julys).

On every map you can hover (or tap, on a phone) to read the value, latitude and longitude of a cell. The colour bar gives the units. Diverging colour scales (blue–white–red) are centred on zero. For net radiation, the cloud effect and the column, blue means energy lost and red energy gained. On anomaly and A − B maps, red means higher than the average (or than B) and blue lower; for outgoing longwave, higher means more energy leaving, so red there is a loss. Sequential scales (dark purple to yellow) are used for plain amounts such as OLR. Grey cells have no value (for example, albedo during the polar night).

Every time series is shaded red during El Niño spells and blue during La Niña spells, from the Niño-3.4 index (section 6): a spell is at least 5 months with the 3-month mean index at or beyond ±0.5 °C.

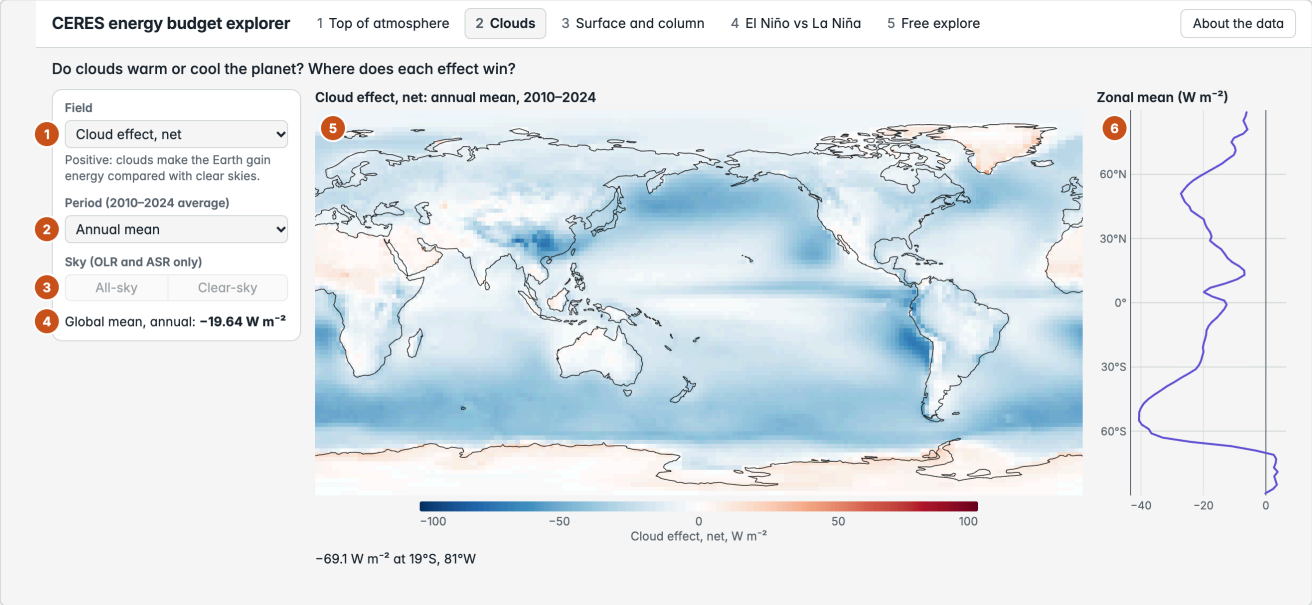
View 1: Top of atmosphere



View 1 with its default settings. The mouse is over the Sahara.

1. **The tabs** for the five views. The address bar keeps the view (`#view=1` to `#view=5`).
2. **About the data**: sources, the sign convention and a link to this handout.
3. **Field**: net radiation, absorbed sunlight, outgoing longwave, planetary albedo or incoming sunlight. The note under the menu says what a positive value means.
4. **Period**: the annual mean or one average month of 2010–2024.
5. **Play the months** steps through the twelve average months.
6. **Global mean** of the map.
7. **The map.** Hover over it to read a value.
8. **Colour bar** with units.
9. **Readout**: the value under the pointer, and where it is.
10. **Zonal mean**: absorbed sunlight and outgoing longwave averaged around each latitude circle, for the period shown.
11. **The global-mean net radiation**, month by month from 2010 to 2024, with its 12-month running mean, its trend, and a readout of both.

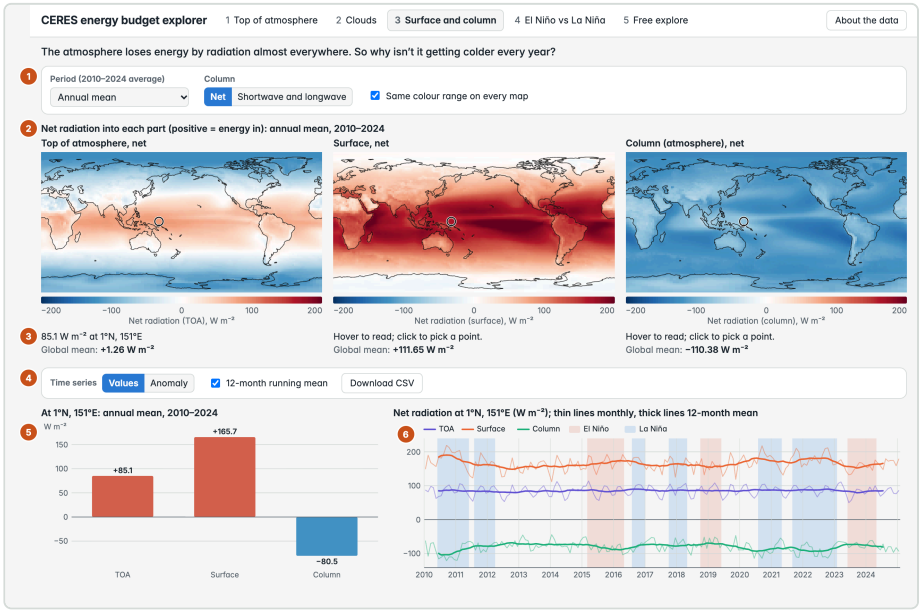
View 2: Clouds



View 2 with its default settings: the net cloud effect at the top of the atmosphere. The mouse is over the ocean off Peru.

1. **Field**: the cloud effect (shortwave, longwave or net) at the top of the atmosphere or at the surface; cloud fraction; cloud-top temperature; and outgoing longwave and absorbed sunlight.
2. **Period**, as in view 1.
3. **Sky**: all-sky or clear-sky. It works only for outgoing longwave and absorbed sunlight, and is greyed out otherwise.
4. **Global mean** of the map.
5. **The map**, with its colour bar and readout below.
6. **Zonal mean** of the field shown.

View 3: Surface and column



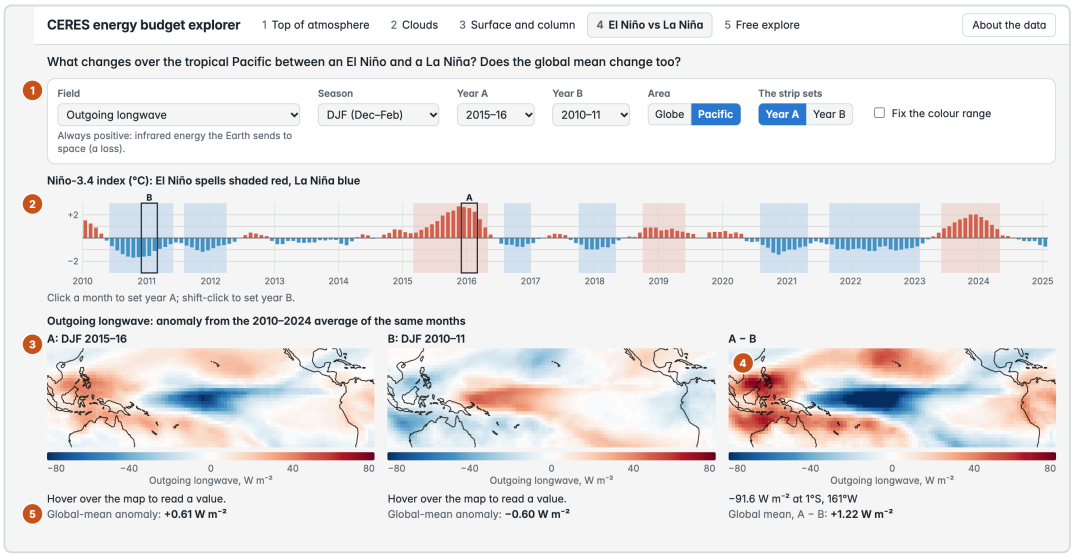
View 3 with the point 1°N, 151°E picked (the circle on each map).

1. **Controls**: the period; **Column**, which shows the column's net radiation or splits it into its **shortwave and longwave** parts; and **Same colour range on every map** (on by default, so the maps can be compared by eye).
2. **Three maps**: net radiation at the top of the atmosphere, at the surface, and for the **column**, the whole depth of the atmosphere between them. The column gains whatever enters at the top and does not reach the surface:

```
column = TOA net − surface net
```

3. **Readout and global mean** under each map. **Click** any map to pick a point.
4. **Time-series controls**: values or anomalies, the 12-month running mean, and **Download CSV**, which saves the series at the picked point (with the Niño-3.4 index) for a spreadsheet or Python.
5. **Bars**: the TOA, surface and column numbers at the picked point, for the period shown.
6. **Time series** of the same numbers at the picked point, 2010 to January 2025.

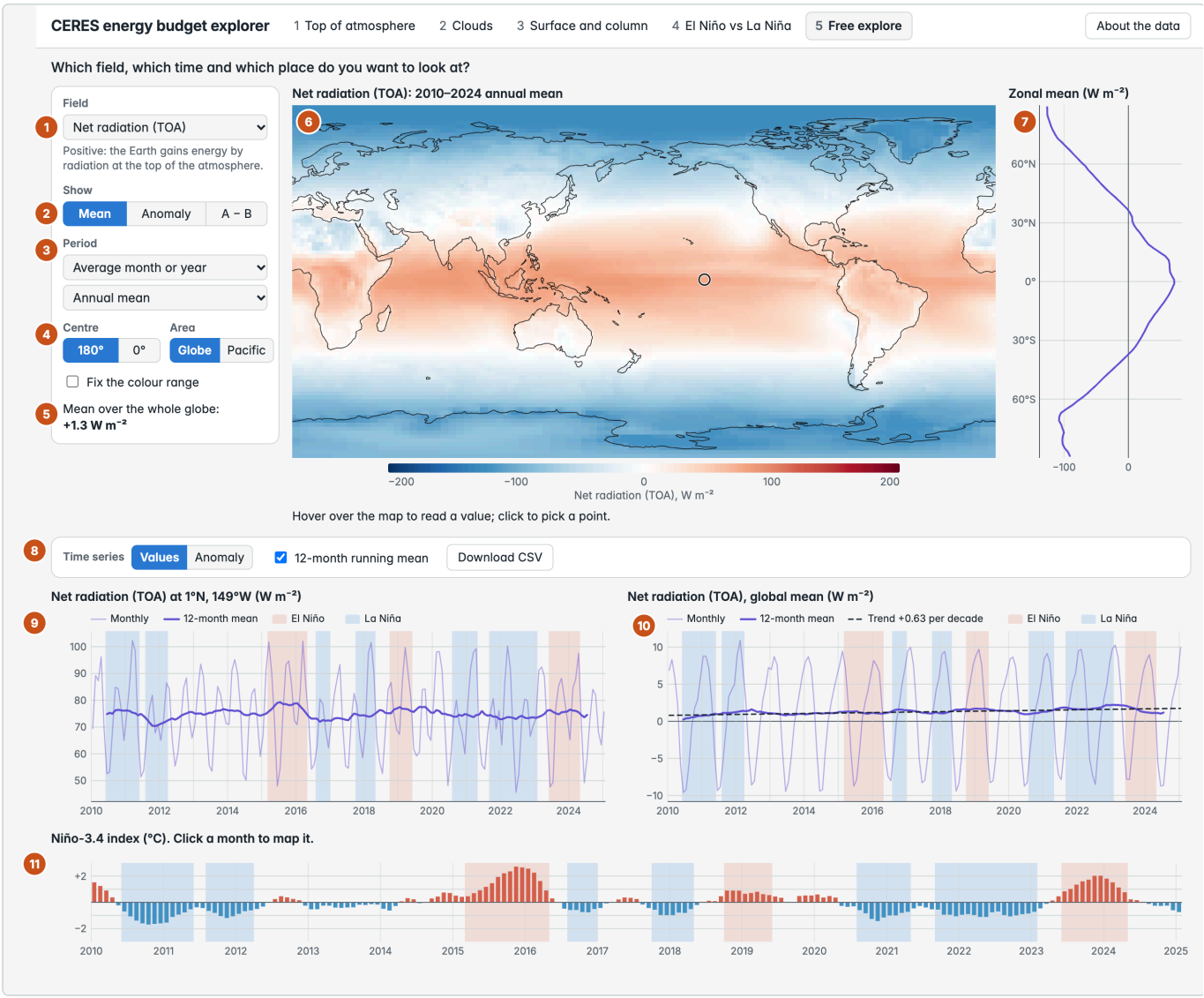
View 4: El Niño vs La Niña



View 4 with its default settings: outgoing longwave, DJF 2015–16 against DJF 2010–11, over the Pacific.

1. **Controls**: any field; the **season**; **Year A** and **Year B** (only complete seasons are offered); **Area** (the whole globe or the tropical Pacific); **The strip sets**, which chooses whether a click on the Niño-3.4 strip sets year A or year B; and **Fix the colour range**.
2. **The Niño-3.4 strip.** Bars are the monthly index; El Niño spells are shaded red and La Niña spells blue. The boxes marked A and B show the months chosen. Click a month to set year A, shift-click to set year B.
3. **Three maps**: the anomaly in A, the anomaly in B, and **A − B**. All three share one colour scale.
4. **A − B** is where the difference between the two events shows best.
5. **Global-mean anomaly** of each map.

View 5: Free explore



View 5 with its default settings. The circle marks the picked point.

1. **Field**: any of the app's fields, grouped by where they are measured.
2. **Show**: **Mean**, **Anomaly**, or **A − B** (period A minus period B).
3. **Period**: an average month or year, one month, one season, or a run of calendar years. In A − B mode a second period, B, appears.
4. **Centre** the map on 180° or 0°, and show the whole **Globe** or zoom to the **Pacific**. Below them, **Fix the colour range** keeps the colours fixed while you change the period.
5. **Mean over the whole globe** of the field shown. It is always the whole-globe mean, even when the map is zoomed to the Pacific.
6. **The map.** Click to pick a point.

7. **Zonal mean** (over the Pacific only, when zoomed).
8. **Time-series controls**, as in view 3. Here **Download CSV** saves the picked point's series, the global-mean series and the Niño-3.4 index.
9. **The series at the picked point**, 2010 to January 2025.
10. **The global-mean series**, with its trend.
11. **The Niño-3.4 strip.** Click a month to map that month.

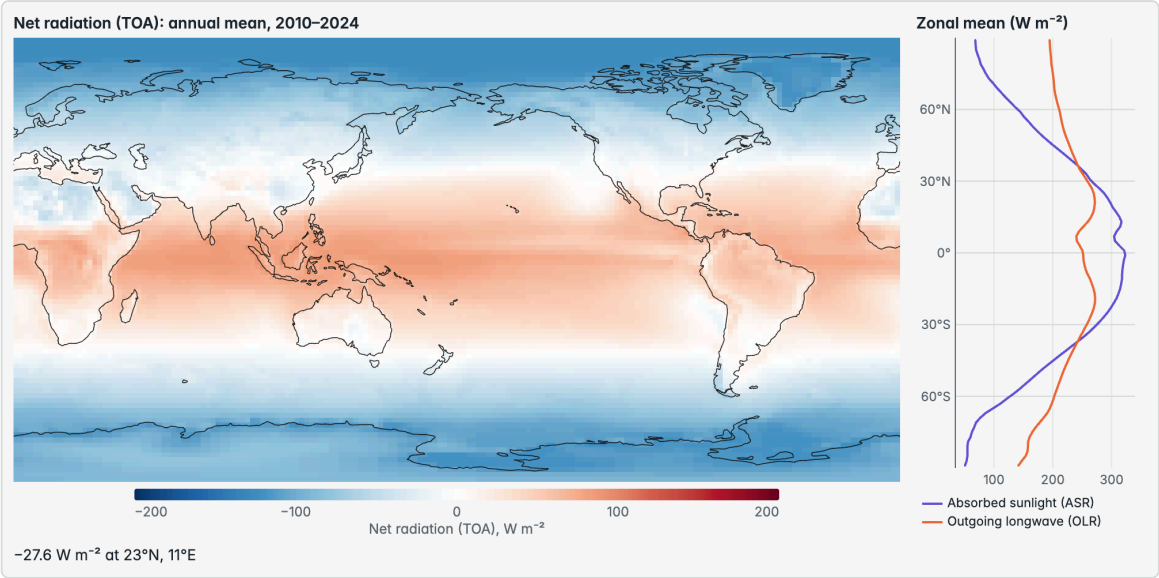
5. Exercises

The numbers below are 2010–2024 averages unless the exercise says otherwise. They are what the app shows, rounded. A point such as "19°S, 81°W" is the centre of a 2° cell. Hover over that spot to read the same value.

Exercise 1. The pattern of the net budget at the top of the atmosphere

Before you look: where on Earth do you expect a net gain of energy by radiation, and where a net loss? Is the boundary between them at the same latitude in both hemispheres? Would a hot desert gain or lose?

In the app: view 1, **Top of atmosphere**, with its defaults: Field **Net radiation (TOA)**, Period **Annual mean**. Hover over the Sahara (23°N, 11°E) and over the western Pacific warm pool (1°N, 151°E). Then switch the Field to **Absorbed sunlight** and to **Outgoing longwave**, and compare.



Annual-mean net radiation, 2010–2024, with the zonal means of absorbed sunlight and outgoing longwave. The readout is for the Sahara.

What to look for

- A **red band of gain** across the tropics and **blue loss** toward both poles. The largest zonal-mean gain is +72 W m^{-2}, at 1°S. The zonal mean near the poles is −126 W m^{-2} at 89°N and −91 W m^{-2} at 89°S.
- In the zonal-mean panel, absorbed sunlight (ASR) and outgoing longwave (OLR) **cross near 37°S and 36°N**. Equatorward of the crossings the Earth gains; poleward of them it loses.
- ASR changes enormously with latitude: about 322 W m^{-2} at the equator against 68 W m^{-2} at 89°N. OLR changes much less: about 251 W m^{-2} at the equator against 195 W m^{-2} at 89°N.
- **The Sahara loses energy** (−28 W m^{-2}), even though it is hot and sunny. It absorbs 255 W m^{-2} of sunlight but emits 283 W m^{-2}. Its bright sand gives it an albedo of 0.34. The warm pool, under heavy cloud, gains +85 W m^{-2} (ASR 315, OLR 230 W m^{-2}, albedo 0.24).

Why

Sunlight arrives almost head-on in the tropics and at a glancing angle near the poles, so the tropics absorb far more of it. Outgoing longwave depends on temperature, and the tropics are only somewhat warmer than the poles, so it varies much less with latitude. The two curves must cross somewhere, and in our climate they cross near 37°S and 36°N.

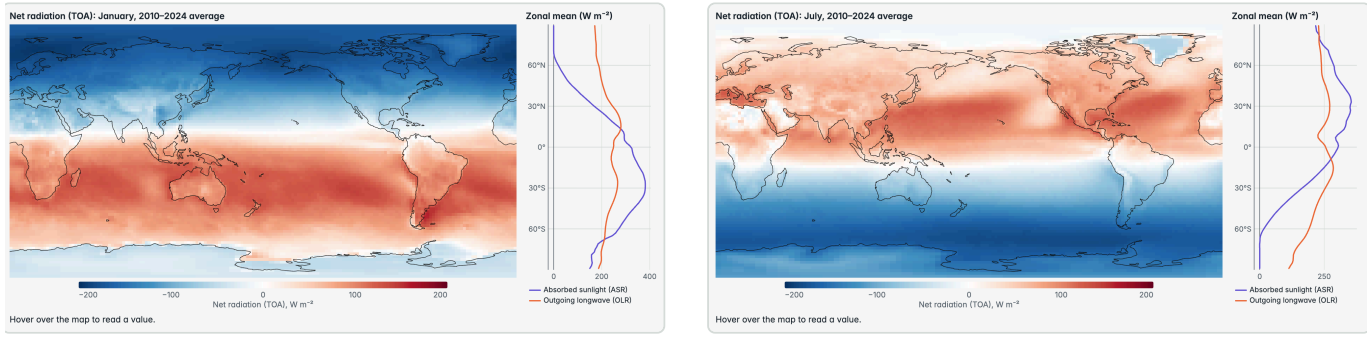
A region that gains energy every year should keep warming, and one that loses should keep cooling. They don't, so **the atmosphere and the ocean must carry energy from the tropics toward the poles.** Adding up the gains equatorward of each latitude, the TOA map implies that the atmosphere and ocean together carry about 5.8 PW (petawatts, 10^{15} W) northward across 36°N, and about 5.7 PW southward across 36°S. Those are the latitudes where the gains stop and the losses begin, so that is where the poleward flow is largest. Weather systems in the atmosphere and currents in the ocean do this carrying.

The Sahara shows that the budget is not just about temperature. Its sand is bright, so it reflects much of the sunlight. Its air is dry and almost cloudless, so its hot surface radiates freely to space. Its atmosphere is kept warm by air that sinks over it and is warmed by compression. That air has brought energy from the rainy tropics.

Exercise 2. Seasons

Before you look: the Earth's tilt moves the Sun north and south through the year. Where will the band of energy gain be in January and in July? In midsummer the Sun never sets at the pole. Could the summer pole take in more energy than it gives off?

In the app: view 1, Field **Net radiation (TOA)**. Click **Play the months** and watch, then pause. Set Period to **January**, then to **July**. In each case, find the latitudes where the zonal-mean absorbed sunlight (purple) is above the outgoing longwave (orange). Then set Field to **Incoming sunlight** for **June** and compare the North Pole with the equator.



Net radiation at the top of the atmosphere in the average January (left) and July (right), 2010–2024.

What to look for

- In **January** the zonal mean shows a gain from 67°S to 13°N. In **July** it runs from 7°S all the way to 81°N. The band of gain follows the Sun into the summer hemisphere and spreads far toward the summer pole.
- The winter pole loses heavily: −171 W m^{-2} at 89°N in January and −113 W m^{-2} at 89°S in July.
- **Near the summer pole the Earth is close to balance.** At 89°N the net is −23 W m^{-2} in June and −11 W m^{-2} in July. At 89°S it is −21 W m^{-2} in December.
- **Incoming sunlight in June** is 517 W m^{-2} at 89°N, but only 386 W m^{-2} at the equator. In December it is 549 W m^{-2} at 89°S against 412 W m^{-2} at the equator.

Why

The incoming sunlight on these maps is a 24-hour average. At the summer solstice the polar Sun is low, but it shines all day and all night. Added up over the day, the pole receives more sunlight than the equator, where it is dark half the time.

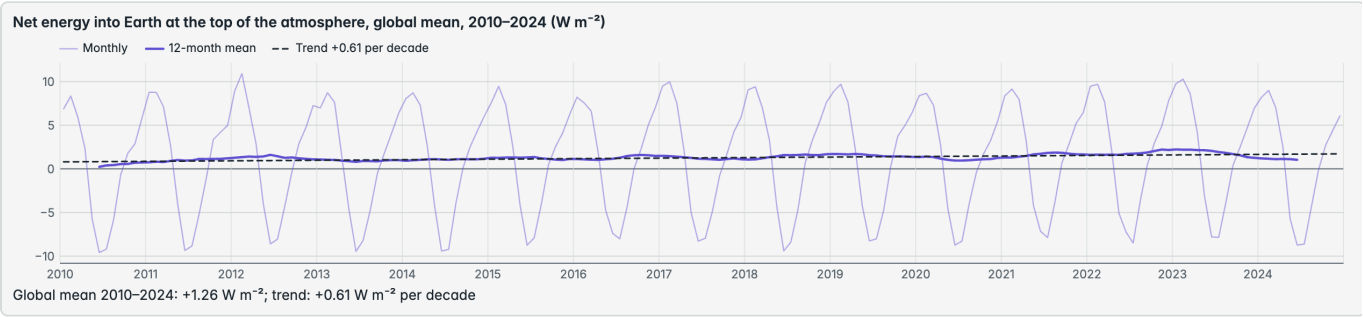
The pole does not absorb most of that sunlight, though. Snow, sea ice and summer clouds make it bright: in June the albedo at 89°N is 0.61, so it absorbs 201 W m^{-2}. That is close to what the pole emits, and the result is a small loss at the pole itself and a **gain over most of the summer high latitudes**, up to 81°N in July. Antarctica is brighter still (an albedo of 0.70 at 89°S in December, absorbing 166 W m^{-2}), so the southern summer gain stops at 67°S, over the ocean around the ice sheet.

In winter the polar night gets no sunlight at all, but the air and surface keep radiating. The energy carried in from lower latitudes keeps them from getting colder and colder. That is why the poleward carrying of energy from exercise 1 is strongest in winter.

Exercise 3. The planetary imbalance

Before you look: if the Earth were in balance, the global-mean net radiation would be zero. Is it? How large is any imbalance compared with the sunlight the Earth absorbs? Has it changed since 2010?

In the app: view 1 with its defaults. Read the **Global mean** in the control panel, then switch the Field to **Absorbed sunlight** and **Outgoing longwave** and read their global means too. Look at the chart below the map and its readout. For a closer look, use view 5 (**Free explore**): Field **Net radiation (TOA)**, Time series **Anomaly**.



Global-mean net radiation, month by month. The thick line is the 12-month running mean; the dashed line is the trend.

What to look for

- The global mean is **+1.26 W m^{-2}**: absorbed sunlight 241.95 minus outgoing longwave 240.69 W m^{-2}. The difference is only 0.5% of the absorbed sunlight.
- The monthly values swing through the year, from about −9 W m^{-2} in the average June to +9 W m^{-2} in the average February, far more than the mean. The 12-month running mean stays positive throughout.
- The readout gives a **trend of +0.61 W m^{-2} per decade**. The 2010–2014 average is +0.93 W m^{-2}; the 2020–2024 average is +1.50 W m^{-2}.
- Among calendar years, 2010 had the smallest imbalance (+0.24 W m^{-2}) and 2023 the largest (+2.06 W m^{-2}).

Why

The Earth is **gaining energy**. Over the whole planet, +1.26 W m^{-2} adds up to about 2.0 × 10^{22} J every year: multiply by the Earth's surface area, 5.1 × 10^{14} m^2, and by the number of seconds in a year. Most of this energy goes into the ocean, which is why the ocean heat content keeps rising. Remember the caveat from section 2: the *size* of the mean imbalance is anchored to the ocean heat measurements, but the *change* since 2010 comes from CERES itself.

The imbalance exists because greenhouse gases keep accumulating. Each increase makes the atmosphere a little more opaque to infrared, so the Earth emits a little less than it absorbs until it warms enough to catch up. The trend shows the gap has grown. Part of that comes from the Earth absorbing more sunlight, as low clouds, snow and sea ice have decreased and some kinds of air pollution have been cleaned up.

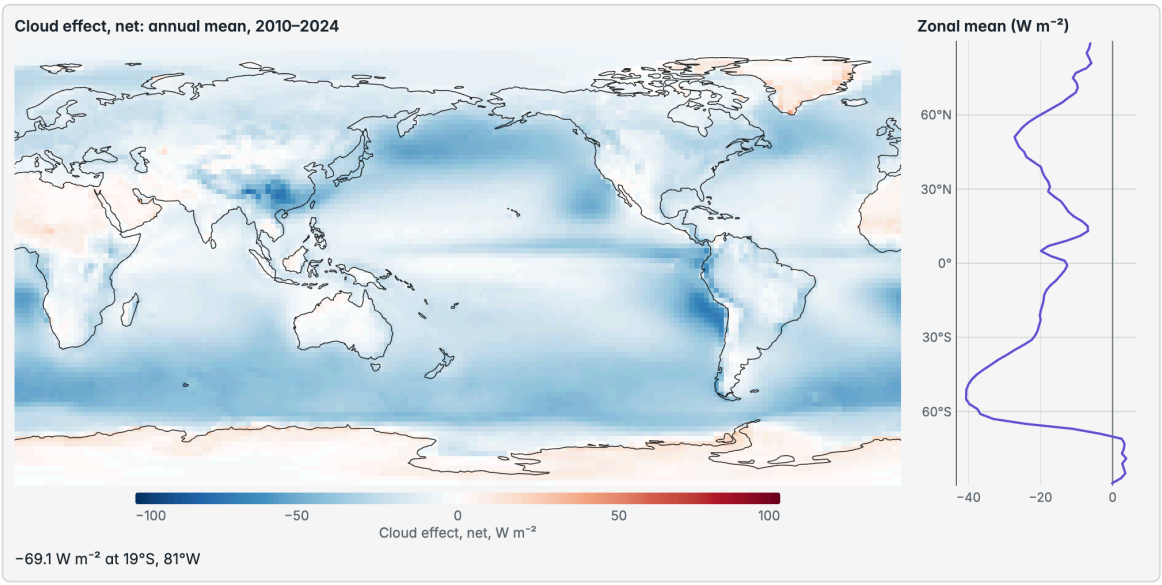
The seasonal swing has nothing to do with this. The Earth is closest to the Sun in early January, and the hemispheres differ (the Southern Hemisphere is mostly ocean). The trend is therefore fitted allowing for a different average for each calendar month, so the seasonal cycle cannot tilt it. In view 5 the same trend reads

+0.63 W m^{-2} per decade, because that series also includes January 2025.

Exercise 4. Clouds cool the planet

Before you look: clouds reflect sunlight back to space, and they also absorb infrared from below and emit less of it to space from their cold tops. Which effect wins, globally? Where do you expect clouds to cool most strongly? Is there anywhere they warm?

In the app: view 2, **Clouds**, with its defaults: Field **Cloud effect, net** (under "Cloud effect (TOA)"), Period **Annual mean**. Read the global mean. Hover over the ocean off Peru (19°S, 81°W), off Namibia (17°S, 7°E) and off California (27°N, 125°W), and over the western Pacific warm pool (1°N, 151°E). Then switch to **Cloud effect, shortwave** and **Cloud effect, longwave** and hover over the same places.



The net cloud effect at the top of the atmosphere, annual mean 2010–2024. The readout is for the ocean off Peru.

What to look for

- Globally the net cloud effect is **−19.6 W m^{-2}**: the shortwave effect is −45.0 W m^{-2} and the longwave effect +25.4 W m^{-2}. Without clouds, the Earth would gain +20.9 W m^{-2} instead of +1.26 W m^{-2}.
- **Dark blue patches off the west coasts** of the Americas and southern Africa:

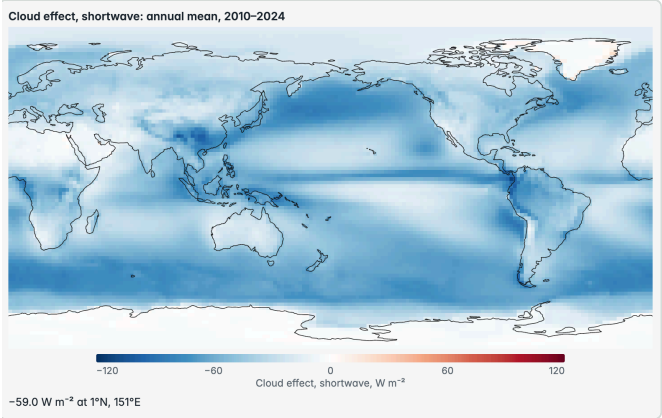
Place	Net	Shortwave	Longwave	Cloud cover	Cloud-top temperature
Off Peru, 19°S, 81°W	−69	−84	+15	87%	279 K
Off Namibia, 17°S, 7°E	−48	−57	+9	78%	284 K
Off California, 27°N, 125°W	−48	−64	+16	78%	276 K
Warm pool, 1°N, 151°E	−9	−59	+50	83%	239 K

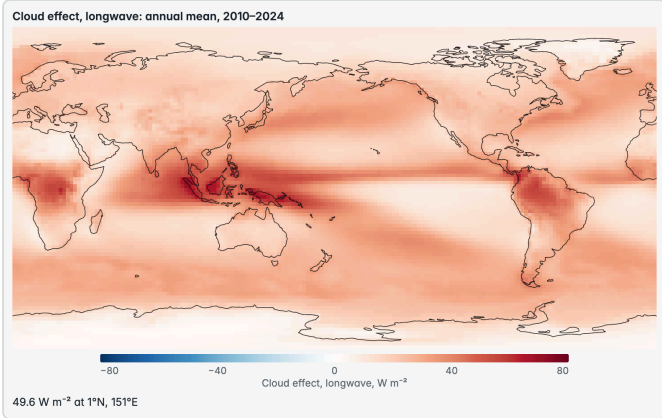
(cloud effects in W m^{-2})

- **Over the warm pool** the two effects are both large and nearly cancel.
- The net effect is positive (red) over Greenland and Antarctica, up to +39 W m^{-2} at 61°N, 45°W.

Why

The **shortwave** cloud effect depends on how much brighter the cloud is than what lies below it. The **longwave** effect depends on how much colder the cloud top is than the surface, because a cold top emits less infrared to space than the warm surface it hides.





Left: the shortwave cloud effect. Right: the longwave cloud effect. Both readouts are for the warm pool, 1°N, 151°E.

The clouds off Peru, Namibia and California are **stratocumulus**: low, flat sheets of cloud over cold ocean water, under air that sinks from above. They are bright and cover most of the sky, so they reflect a great deal of sunlight. Their tops are low and nearly as warm as the sea (279 K off Peru), so they hide very little infrared. **The shortwave wins** and they cool strongly.

Over the warm pool, the clouds are **deep convective towers** and the anvils that spread from them. They also reflect a lot of sunlight (a shortwave effect of −59 W m^{-2}, though less than the −84 W m^{-2} off Peru), but their tops are high and cold (239 K, about −34 °C), so they trap a lot of infrared. **The two effects nearly cancel.**

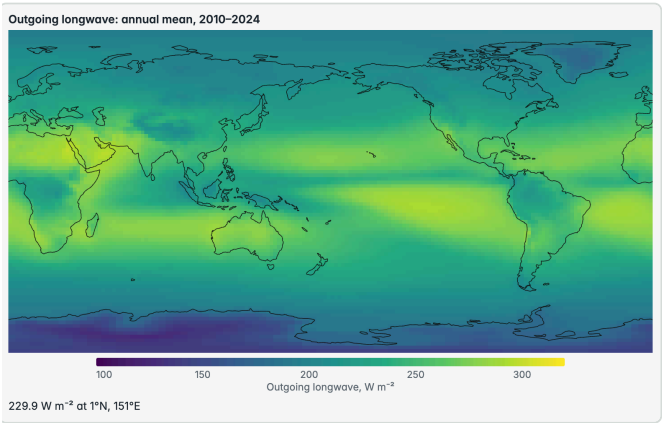
Over Greenland and Antarctica the surface is snow, already as bright as a cloud, so clouds add little reflection; but they still trap infrared, so there they warm.

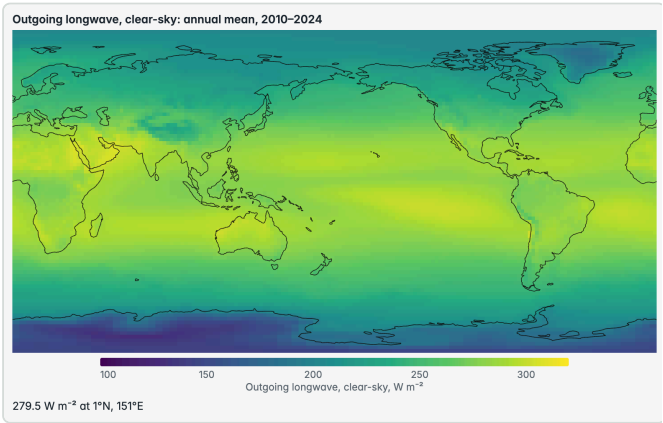
Across the whole globe, the cooling wins, and by a lot: the cloud effect is many times the planet's imbalance. This is why a small change in clouds as the climate warms matters so much.

Exercise 5. Clear-sky and all-sky outgoing longwave: the warm pool and the Sahara

Before you look: the warm pool's sea surface is about as warm as any on Earth, and the Sahara is hot. Which of the two sends more infrared to space? Would the answer change if you could remove the clouds?

In the app: view 2. Field **Outgoing longwave**, Sky **All-sky**, then **Clear-sky**. Hover over the warm pool (1°N, 151°E) and the Sahara (23°N, 11°E) each time. Then look at **Cloud fraction** and **Cloud-top temperature** at the same two places.





Outgoing longwave with the clouds (left) and without them (right). Both readouts are for the warm pool.

What to look for

- **Warm pool:** 230 W m^{-2} with clouds, 280 W m^{-2} without them. The clouds hold back +50 W m^{-2}. Cloud covers 83% of the sky there, with tops at 239 K (−34 °C).

- **Sahara:** 283 W m^{-2} with clouds, 293 W m^{-2} without them. The difference is only +10 W m^{-2}, because cloud covers just 17% of the sky.
- With the clouds removed, the two differ by only about 13 W m^{-2}. With the clouds, the warm pool sends out about 53 W m^{-2} less.
- On the all-sky map, the lowest tropical values (darkest) mark the rainy places: the warm pool, the Amazon, central Africa, and a thin line north of the equator across the Pacific and Atlantic.

Why

A surface or a cloud emits about σT^4, where σ is the Stefan–Boltzmann constant and T its temperature in kelvin. A cloud top at 239 K emits about 186 W m^{-2}, far less than the warm sea below it. Where deep clouds cover most of the sky, as over the warm pool, the planet "shows" space mostly its cold cloud tops, and the outgoing longwave drops.

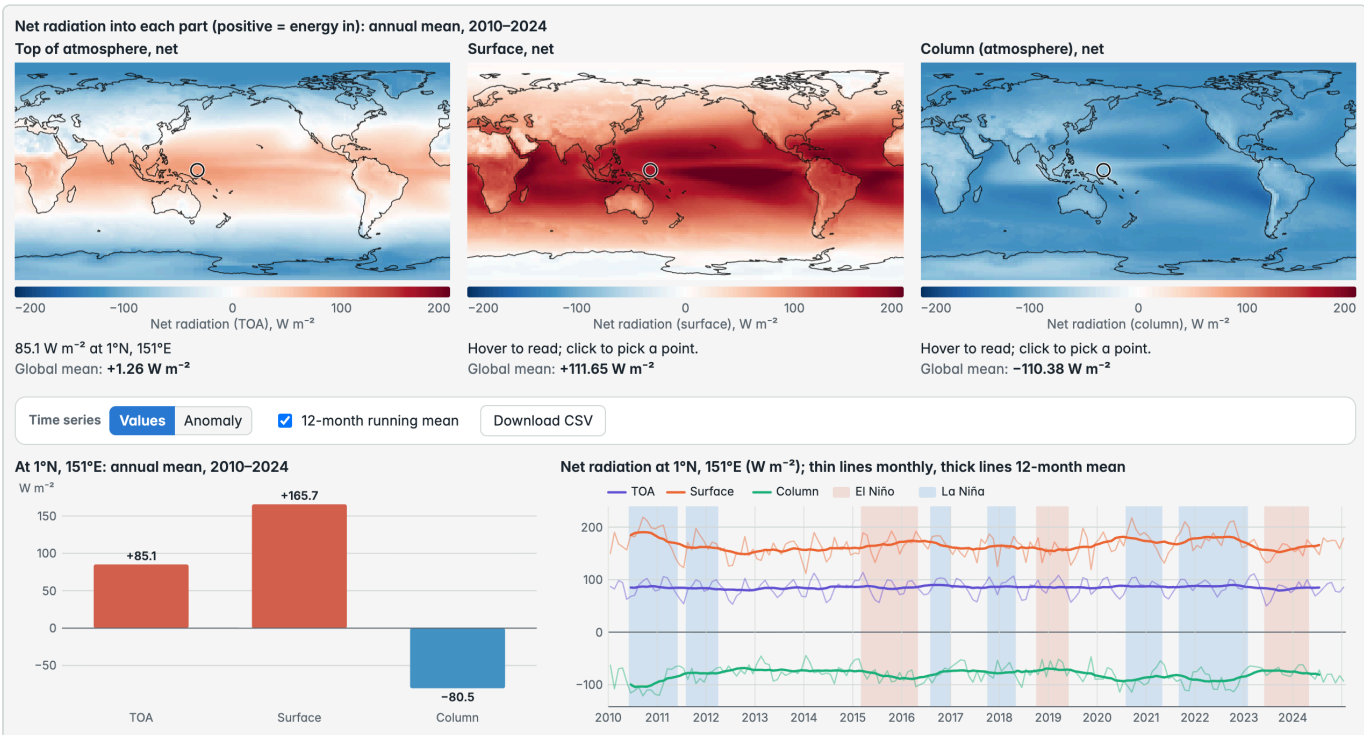
Even without clouds, the warm pool emits less than the Sahara, although its surface is just as warm. Its air is full of water vapour, a greenhouse gas, which absorbs the surface's infrared and re-emits it from higher, colder levels. The Sahara's air is dry, so more of its hot surface's infrared escapes directly.

This is why weather forecasters use low outgoing longwave to find tropical rain: low OLR in the tropics means high, cold cloud tops, and those come from deep, raining convection. You will use this in exercise 7.

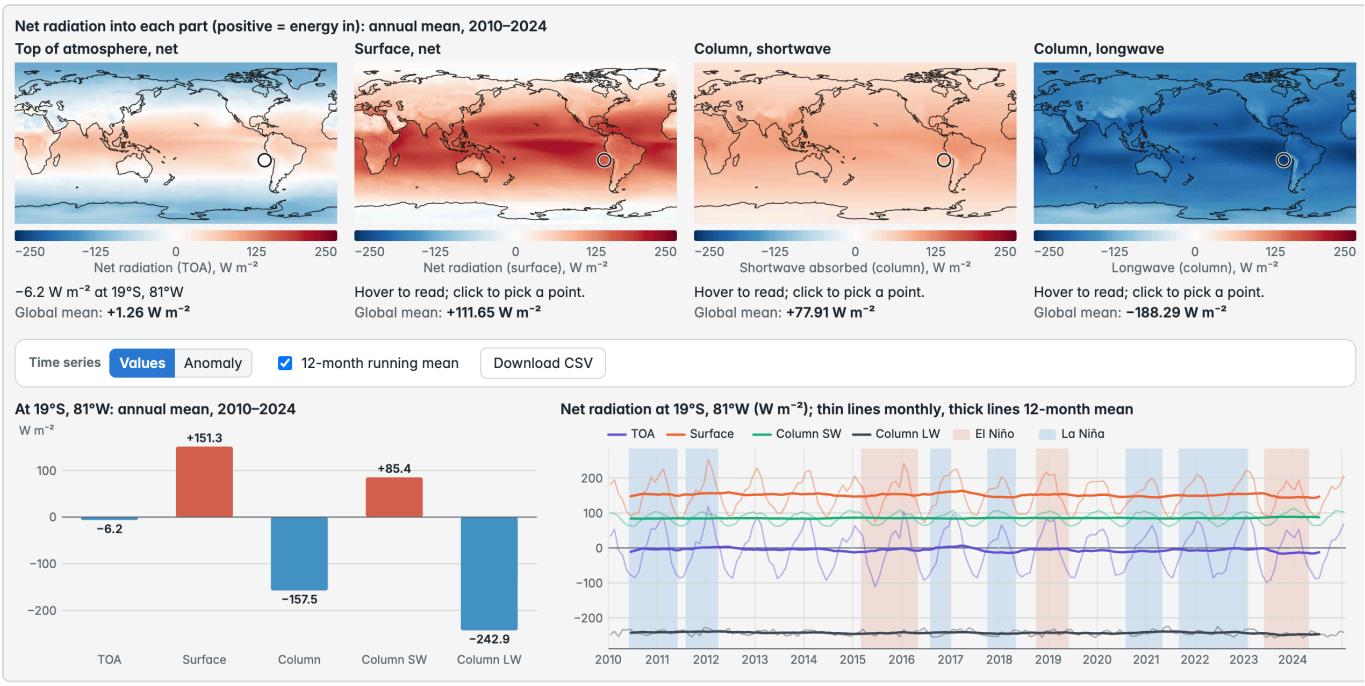
Exercise 6. The column puzzle

Before you look: the atmosphere absorbs some sunlight, absorbs infrared from the surface, and emits infrared both up to space and down to the surface. Does it gain or lose energy by radiation overall? If it loses, why doesn't it get colder every year?

In the app: view 3, **Surface and column**, with its defaults. Read the three global means. Click the warm pool (1°N, 151°E), then the ocean off Peru (19°S, 81°W), then the Sahara (23°N, 11°E), and read the bars. Then set **Column** to **Shortwave and longwave**.



Net radiation at the top of the atmosphere, at the surface and in the column. The bars are for the warm pool.



The column split into shortwave and longwave, with the ocean off Peru picked.

What to look for

- The global means: **+1.26 W m^{-2}** at the top of the atmosphere, **+111.6 W m^{-2}** at the surface, and **−110.4 W m^{-2}** for the column.
- **The column loses energy by radiation everywhere**, in every one of the 16200 cells of the annual-mean map. The loss ranges from −54 W m^{-2} (3°N, 97°E) to −158 W m^{-2} (23°S, 75°W).
- Split into parts, the column **absorbs** +77.9 W m^{-2} of sunlight but **loses** −188.3 W m^{-2} in the longwave.
- **Over the oceans**, the loss is smaller in the rainy places (−81 W m^{-2} over the warm pool) than over the dry subtropical oceans (−157 W m^{-2} off Peru). Over the Sahara, dry as it is, the loss is only −69 W m^{-2}.
- At the surface the warm pool gains +166 W m^{-2} by radiation, and the ocean off Peru +151 W m^{-2}. The Sahara's surface gains only +42 W m^{-2}.

Why

The atmosphere really does lose about 110 W m^{-2} by radiation. With nothing else heating it, the whole atmosphere would cool by about 0.9 °C every day. It doesn't, so something other than radiation must heat it, by the same amount. The surface tells you where to look: it gains +111.6 W m^{-2} by radiation, yet it doesn't keep warming either. **The surface hands its excess energy to the atmosphere, without radiation**, in two ways.

1. **Latent heat.** Water evaporates from the surface, mostly from the ocean. Evaporation takes energy: it is why sweat cools you. The water vapour carries that energy up into the atmosphere. When the vapour condenses into cloud droplets and falls as rain, the energy is released into the air.
2. **Sensible heat.** Air touching a warmer surface is heated by it, and turbulence carries the warmed air upward.

Estimates of the global means, from many observations put together, are roughly **80 W m^{-2} of latent heat** and roughly **20 W m^{-2} of sensible heat**. Trenberth and colleagues (2009) gave 80 and 17 W m^{-2}; Stephens and colleagues (2012) gave 88 and 24 W m^{-2}. Their sums, 97 and 112 W m^{-2}, bracket the column's radiative loss of 110 W m^{-2}. These estimates, and the computed surface fluxes in EBAF, are each uncertain by several W m^{-2}, so this is as close an agreement as we can expect.

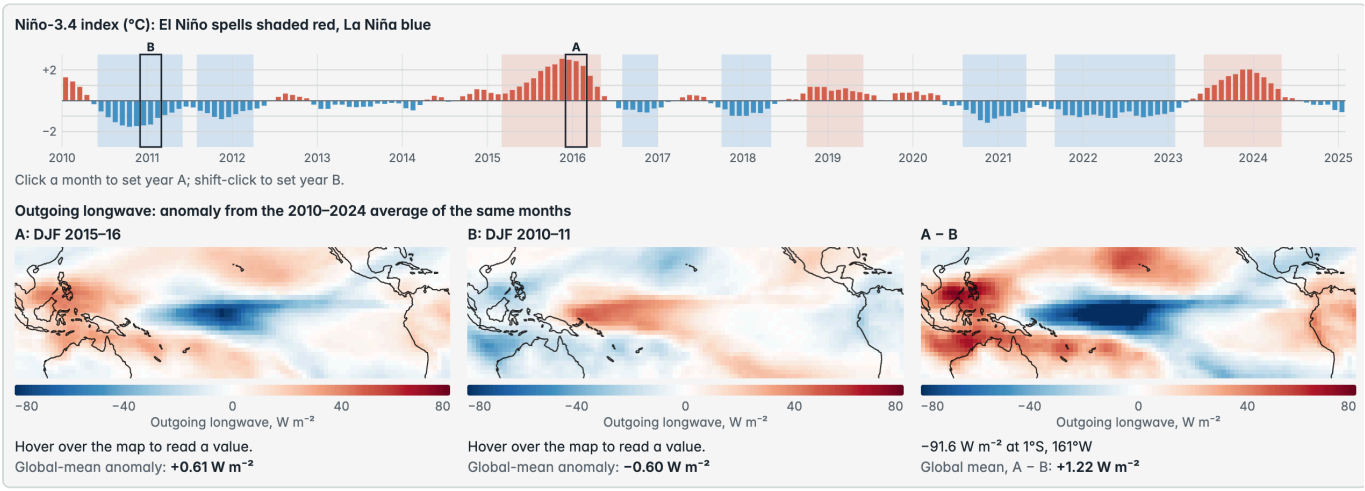
Rain is the latent heat you can see. Condensing enough water to give 1 mm of rain a day releases about 29 W m^{-2}. So 80 W m^{-2} of latent heating means that, averaged over the globe, about 2.8 mm of rain falls each day. This is how the ocean hands energy to the atmosphere: sunlight warms the sea surface, the sea evaporates, and the energy is released in the air, in clouds and rain.

But the map has a twist. If the rain balanced the radiation in each place, the rainiest places should show the biggest radiative loss. Over the oceans they show the opposite. The warm pool loses only −81 W m^{-2}: its high, thick clouds trap infrared inside the column (the column's cloud effect there is +40 W m^{-2}). The biggest losses are over dry subtropical oceans like the one off Peru (−157 W m^{-2}), where it hardly ever rains. (Dry land is different. The Sahara's ground gets very hot and loses −130 W m^{-2} as infrared, much of which the air above absorbs, and the hot ground also heats the air directly as sensible heat. A cool ocean surface can do neither, so the Sahara's column loses only −69 W m^{-2}.) So rain cannot balance the budget place by place. The atmosphere also **moves energy sideways**. In the rain belts, condensation releases far more heat than the column loses by radiation, and the air rises. That air spreads toward the subtropics and sinks there, warming by compression as it sinks. The sinking warms the dry columns that lose the most. The water vapour that feeds the rain belts was mostly evaporated over the warm subtropical oceans under the trade winds, and the trade winds carried it toward the equator. (Off Peru the sea also gains a lot by radiation, +151 W m^{-2}, but much of that warms cold water that wells up from below along the coast, rather than evaporating it: another place where the ocean stores and moves the energy.) This circulation, rising in the deep tropics and sinking in the subtropics, is the **Hadley circulation**. Its east–west cousin along the equatorial Pacific, the **Walker circulation**, is the one El Niño disturbs (exercise 7).

Exercise 7. El Niño vs La Niña

Before you look: during El Niño, the warmest water spreads east from the western Pacific into the central and eastern Pacific. Where do you expect more deep cloud and rain during El Niño than during La Niña, and where less? What will that do to the outgoing longwave there? And to the net radiation?

In the app: view 4, **El Niño vs La Niña**, with its defaults: Field **Outgoing longwave**, Season **DJF**, Year A **2015–16** (a strong El Niño), Year B **2010–11** (a strong La Niña), Area **Pacific**. Hover over the A − B map at 1°S, 161°W and at 1°S, 119°E. Then change the Field to **Cloud effect, shortwave**, **Cloud effect, longwave** and **Net radiation (TOA)**. Finally set Year A to **2023–24** and Year B to **2020–21**, and repeat.

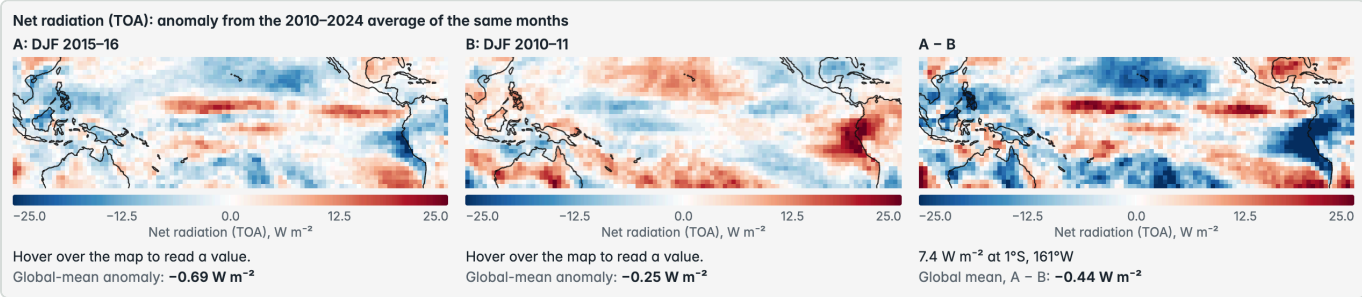


Outgoing longwave anomalies over the Pacific in DJF 2015–16 (El Niño), DJF 2010–11 (La Niña), and the difference. The readout is for 1°S, 161°W.

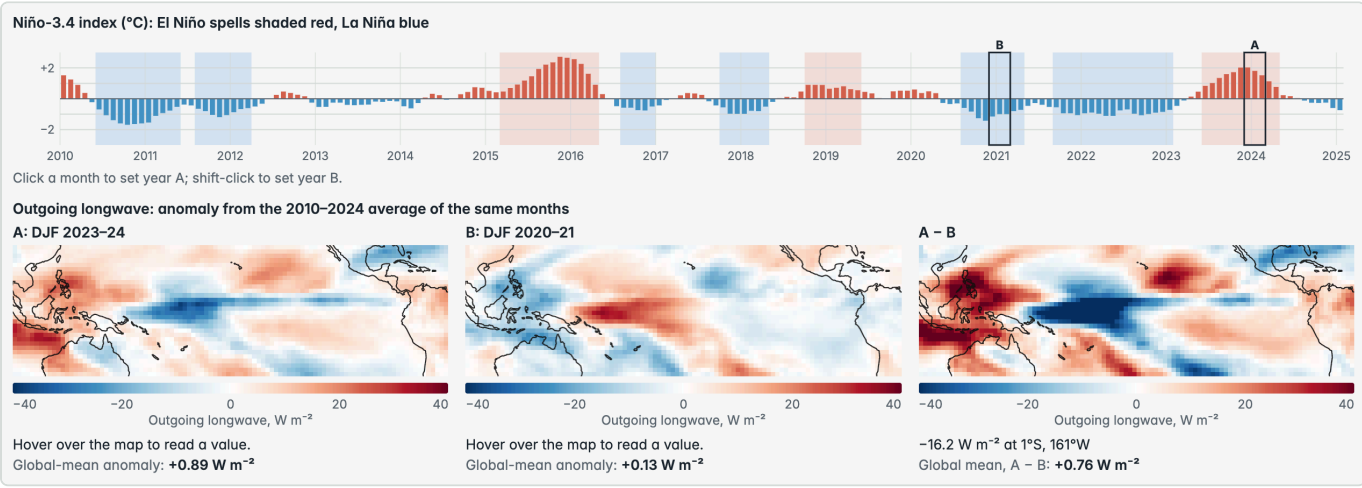
What to look for

- In DJF the Niño-3.4 index averaged +2.5 °C in 2015–16 and −1.4 °C in 2010–11.
- **A − B for outgoing longwave** is a **dipole**. It is strongly negative along the equator in the central Pacific: −92 W m^{-2} at 1°S, 161°W, and −104 W m^{-2} at its strongest, at 1°N, 173°W. It is positive over the maritime continent (Indonesia and the seas around it): +30 W m^{-2} at 1°S, 119°E.
- At 1°S, 161°W the **cloud effects** change a lot but in opposite directions: shortwave −81, longwave +71 W m^{-2}. The net cloud effect changes by only −10 W m^{-2}.

- **Net radiation** changes far less than outgoing longwave: +7 W m^{-2} at 1°S, 161°W and +11 W m^{-2} at 1°S, 119°E. Absorbed sunlight changes almost as much as OLR (−84 W m^{-2} at 1°S, 161°W), and the two nearly cancel.
- **2023–24 minus 2020–21** (Niño-3.4 +1.8 °C against −1.1 °C) shows the same dipole, but weaker and further west: the largest drop in OLR is −67 W m^{-2}, at 3°N, 177°E, while at 1°S, 161°W it is only −16 W m^{-2}.



The same two seasons for net radiation at the top of the atmosphere. Note the much smaller colour range than for outgoing longwave.



The 2023–24 El Niño against the 2020–21 La Niña.

Why

Deep convection, the tall rain clouds of exercise 5, sits over the warmest water. Normally that is the western Pacific warm pool. During El Niño the warm water and the convection shift east to the central Pacific, and during La Niña they are pushed further west than usual. So El Niño minus La Niña gives **more high, cold cloud** in the central Pacific (much lower OLR) and **less** over the maritime continent (higher OLR). The Walker circulation weakens and shifts with them.

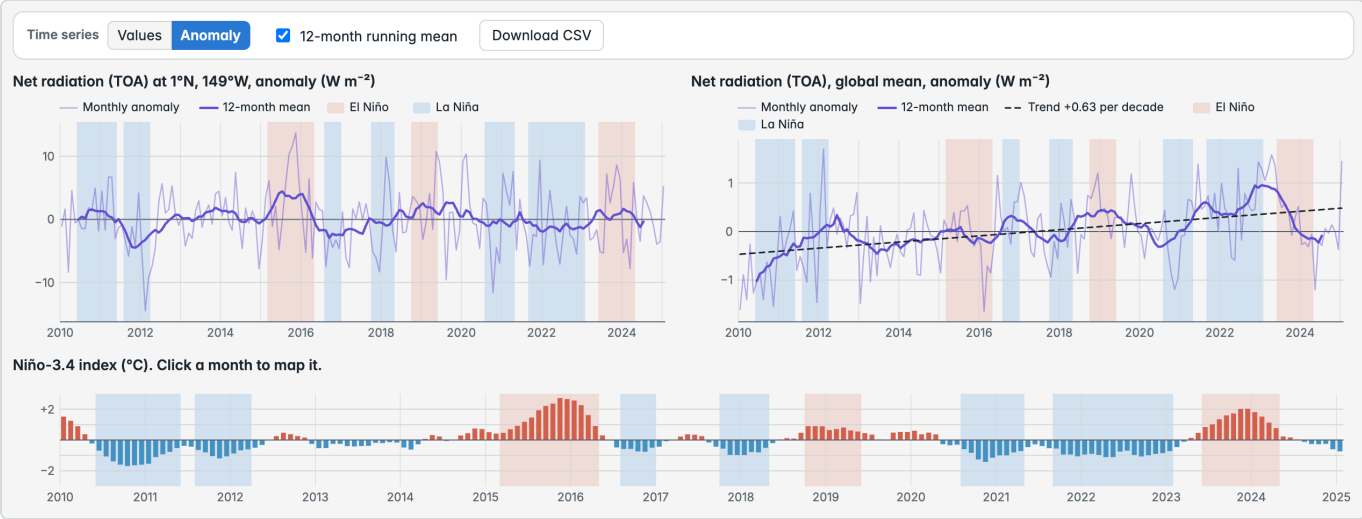
The net radiation hardly changes, for the reason you found in exercise 4: deep convective clouds reflect about as much sunlight as the infrared they trap. Moving the clouds east moves both effects together. **El Niño rearranges the clouds and the rain far more than it changes the energy gained at the top of the atmosphere.** Its effect on the energy budget is mainly at the surface and inside the ocean and atmosphere: the ocean releases heat to the atmosphere in the central and eastern Pacific.

No two El Niños are alike. In 2023–24 the warming peaked further west, and the cloud changes followed it.

Exercise 8. Does El Niño change the *global* budget?

Before you look: in exercise 7 the Pacific changed a lot. Will the global-mean net radiation be higher or lower during El Niño? Will the change happen at the same time as the Niño-3.4 peak, or later?

In the app: view 4 again, with the defaults: read the global-mean anomalies under the maps, first for **Outgoing longwave** and then for **Net radiation (TOA)**. Then go to view 5, **Free explore**: Field **Net radiation (TOA)**, Time series **Anomaly**. Compare the global-mean anomaly with the Niño-3.4 strip below it. If you want to go further, click **Download CSV**. The file has columns `time`, the series at the picked point, `toa_net_anom_global` and `nino34_anom`. In a spreadsheet or Python, compute the correlation between `nino34_anom` and `toa_net_anom_global`, then shift the global column one month later at a time and recompute it. Repeat with Field **Outgoing longwave**.



Free explore, net radiation, anomaly series. Right: the global mean. Bottom: the Niño-3.4 index.

What to look for

- In view 4, DJF 2015–16 against DJF 2010–11: the global-mean **outgoing longwave** anomaly is +0.61 W m^{-2} in A and −0.60 W m^{-2} in B, so A − B is +1.22 W m^{-2}. The global-mean **net radiation** anomaly is −0.69 W m^{-2} in A and −0.25 W m^{-2} in B, so A − B is −0.44 W m^{-2}.
- For 2023–24 against 2020–21, the outgoing longwave A − B is +0.76 W m^{-2}, but the net radiation A − B is **+0.39 W m^{-2}**: the opposite sign, because the absorbed sunlight rose even more (+1.15 W m^{-2}).
- The monthly global net anomaly has a standard deviation of 0.7 W m^{-2} and runs from −1.7 to +1.7 W m^{-2}. The Niño-3.4 index explains little of that:

Correlation of Niño-3.4 with the global anomaly of…	Same month	Strongest
Net radiation (TOA)	+0.03	−0.23, 6 months later
Outgoing longwave	+0.32	+0.41, 2 months later
Absorbed sunlight	+0.27	+0.27, same month

- Averaged over all El Niño months, the global outgoing longwave anomaly is +0.39 W m^{-2}; over all La Niña months, −0.20 W m^{-2}.

Why

During El Niño the ocean releases heat into the atmosphere over the central and eastern Pacific. The atmosphere warms, and a warmer atmosphere emits more infrared to space: the global OLR rises, peaking a couple of months after Niño-3.4 does. On its own, that would make the Earth lose energy during El Niño. But the absorbed sunlight changes too, because the clouds rearrange (exercise 7), and in these years it rose with the OLR. The two largely cancel. Only about half a year after the Niño-3.4 peak does the global net show a small loss (the correlation of −0.23). La Niña does the reverse: the ocean stores heat and the planet emits a little less.

So the answer is: **a little, and late.** El Niño mainly moves energy between the ocean and the atmosphere, and from place to place, rather than adding energy to the planet or removing it. Months of El Niño and La Niña make the global-mean surface temperature jump about, because heat moves out of or into the ocean. But the long-term imbalance of exercise 3 is set by the greenhouse gases and the planet's reflectivity, and El Niño only adds wiggles to it. 2023 shows how much else matters. An El Niño year would be expected to lose energy, yet in 2023 the Earth gained more energy than in any other year of the record, mostly because it absorbed more sunlight.

6. Data credit and further reading

Data in the app

- CERES radiation: NASA Langley Research Center, CERES Science Team. *CERES Energy Balanced and Filled (EBAF) TOA and Surface Monthly means data in netCDF, Edition 4.2.1.* NASA Langley Atmospheric Science Data Center. doi:10.5067/TERRA-AQUA-NOAA20/CERES/EBAF_L3B004.2.1. Months used: January 2010 to January 2025; the app averages the 1° grid onto a 2° grid.
- Niño-3.4 index: NOAA Climate Prediction Center, monthly Niño-3.4 sea-surface temperature anomaly from ERSSTv5, relative to the 1991–2020 base period, www.cpc.ncep.noaa.gov/data/indices/ersst5.nino.mth.91-20.ascii. The sea-surface temperature data set is described by Huang, B., and co-authors (2017), Extended Reconstructed Sea Surface Temperature, Version 5 (ERSSTv5), *Journal of Climate* 30, 8179–8205, doi:10.1175/JCLI-D-16-0836.1.
- Coastlines: Natural Earth, 1:110 million coastline, public domain, naturalearthdata.com.

The CERES instruments and EBAF

- Wielicki, B. A., and co-authors (1996). Clouds and the Earth's Radiant Energy System (CERES): An Earth Observing System experiment. *Bulletin of the American Meteorological Society* 77, 853–868. doi:10.1175/1520-0477(1996)077<0853:CATERE>2.0.CO;2
- Loeb, N. G., and co-authors (2018). Clouds and the Earth's Radiant Energy System (CERES) Energy Balanced and Filled (EBAF) Top-of-Atmosphere (TOA) Edition-4.0 data product. *Journal of Climate* 31, 895–918. doi:10.1175/JCLI-D-17-0208.1. How the TOA fluxes are made and energy-balanced.
- Kato, S., and co-authors (2018). Surface irradiances of Edition 4.0 Clouds and the Earth's Radiant Energy System (CERES) Energy Balanced and Filled (EBAF) data product. *Journal of Climate* 31, 4501–4527. doi:10.1175/JCLI-D-17-0523.1. How the surface fluxes are computed, and how uncertain they are.
- Loeb, N. G., and co-authors (2020). Toward a consistent definition between satellite and model clear-sky radiative fluxes. *Journal of Climate* 33, 61–75. doi:10.1175/JCLI-D-19-0381.1. The "total region" clear-sky fluxes used in this app.

The surface energy budget (exercise 6)

- Trenberth, K. E., J. T. Fasullo and J. Kiehl (2009). Earth's global energy budget. *Bulletin of the American Meteorological Society* 90, 311–323. doi:10.1175/2008BAMS2634.1. Latent heat 80 W m^{-2}, sensible heat 17 W m^{-2}.
- Stephens, G. L., and co-authors (2012). An update on Earth's energy balance in light of the latest global observations. *Nature Geoscience* 5, 691–696. doi:10.1038/ngeo1580. Latent heat 88 ± 10 W m^{-2}, sensible heat 24 ± 7 W m^{-2}.

The growing imbalance (exercise 3)

- Loeb, N. G., and co-authors (2021). Satellite and ocean data reveal marked increase in Earth's heating rate. *Geophysical Research Letters* 48, e2021GL093047. doi:10.1029/2021GL093047.

Every number in sections 1–5 that comes from the data was computed by `docs/handout_numbers.mjs`*, which reads the app's own data files with the app's own code; the results are in* `docs/handout_numbers.txt`*. The screenshots were made with* `docs/handout_shots.mjs`*.*